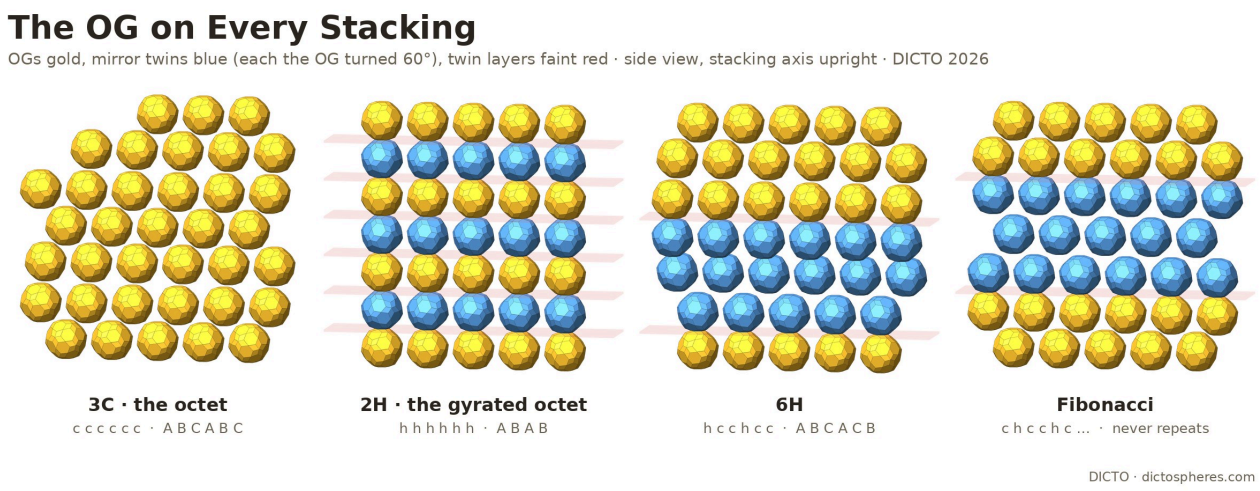
The DICTO OG Family on Every Stacking

Twins, polytypes and a Fibonacci crystal: two general rules for the Ogdoecontahedron and its family in the tetrahedral–octahedral honeycomb stacked in any order, with exact kits for the classical polytypes and for a stacking that never repeats

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Version 1 · 20 October 2026 · DICTO DOI: reserved on publication · Companion to The DICTO Ogdoecontahedron on the Gyrated Octet (DICTO DOI: 10.5281/zenodo.23484012) · Licence CC BY-SA 4.0 · Scripts, exact checks and data: https://github.com/DICTOR-Master/kaleidohedra/tree/18627b1/studies/2026-10-20-og-polytypes (research branch, commit 18627b1; also attached to this record)
The Ogdoecontahedron in the gyrated octet, the family of solids and their names are DICTO's [3]; taking the family to every stacking was proposed by Claude and adopted by DICTO, and the rules, the polytypes and the Fibonacci stacking are joint work. Computation, checking and drafting with Claude (Anthropic).



Abstract. The tetrahedral–octahedral honeycomb falls into layers, and each layer can be laid on the one below in two ways: as in the octet truss (c) or as its mirror image in the shared plane (h). Any word in c and h is a stacking, exactly as for close-packed crystals, whose polytypes 3C, 2H, 4H, 6H, 9R, 15R … are words of this kind. The DICTO Ogdoecontahedron (OG) sits exactly in the octahedra of the octet [2] and of the gyrated octet [3]; here we place it, and the solids of its family, in **every** stacking. Two rules decide it. **The twin rule**: a solid that is centre-symmetric and has a three-fold axis at right angles to the layers has as its mirror image in a layer plane the same solid turned 60°; unless it also has a mirror at right angles to the axis, this is a true twin. **The fit rule**: a solid that lies in its own layer and does not overlap its copies within the layer fits every stacking, whatever the word. Exactly in $\mathbb{Q}(\sqrt{5})$: the OG, the DICTO OTT and the OCD Stella lie in their octahedra and fit every stacking with one cell kit; the **DICTO Belted Triacontahedron (DBT)** leaves its octahedron but not its layer, so it too fits every stacking (shown in [3] only for the gyrated octet); the golden envelope it was cut from fits the octet but overlaps at every h layer by exactly the volume of the two tips the DBT removes. The shares of space do not depend on the stacking: OG $1025/114 - 145\sqrt{5}/38 \approx 45.89\%$, OTT $176/3 - 26\sqrt{5} \approx 52.89\%$, DBT $493/6 - 73\sqrt{5}/2 \approx 55.02\%$. For the OG we give the kit per cell for 3C, 2H, 4H, 6H, 9R and 15R, and for the **Fibonacci stacking** (c → ch,

> h → c), which never repeats: there exactly (3 − √5)/2 of the layers are twin layers, and a cell needs 5 + √5 caps, (3 − √5)/2 twin caps, √5 − 1 tetrahedra and (3 − √5)/2 bipyramids, all exact.

1. Stackings of the honeycomb

Layers. Frame as in [2, 3]: the octahedron |x| + |y| + |z| ≤ R, R = 2φ/3; a cell of the honeycomb (one octahedron, two tetrahedra) has volume (32 + 16√5)/27. Cut at right angles to the three-fold axis (1,1,1), the honeycomb falls into layers one octahedron thick, the slabs (2k − 1)R ≤ x + y + z ≤ (2k + 1)R. The planes between layers hold the honeycomb's vertices, which are the centres of a close packing of spheres; the layers of the honeycomb lie between the layers of that packing.

Two steps. On a layer the next one lies either as in the octet truss (the step c, a translation) or as the mirror image of the layer in the plane they share (the step h). Across a c plane an octahedron meets tetrahedra; across an h plane octahedron meets octahedron and tetrahedron meets tetrahedron. Any word in c and h is a stacking: all c is the octet (fcc, 3C), all h the gyrated octet (hcp, 2H) [4, 5]. These are the h and c of Jagodzinski's notation for close packings [6] (c is often written k), and periodic words give the polytypes named by Ramsdell [7] and the IUCr [8].

Built and read back (exact). polytype_exact.py builds any word layer by layer with rational maps (a mirrored layer takes its next step downward in the reference frame and is then mirrored up), checks that layer k lies at x + y + z = 2kR, and reads the word back from the geometry: above an octahedron's top face lies an octahedron exactly at the h planes. The A/B/C sequence of the packing follows from the word (c keeps the turn A → B → C, h reverses it) and gives the Ramsdell symbols in Table 2.

2. The twin rule

Rule. Let a solid S, centred at the origin, be centre-symmetric (S = −S) and have a 120° turn about the axis n at right angles to the layers. Then its mirror image in a plane at right angles to n is S turned 60° about n.

Proof. The mirror M in the plane through the centre is the inversion composed with the half turn H about n: M = −H. With S = −S, M(S) = H(S). The half turn is three 60° turns, $H = T^{60} \circ T^{60} \circ T^{60} = T^{60} \circ T^{120}$, and $T^{120}(S) = S$, so $H(S) = T^{60}(S)$. A mirror in a parallel plane differs by a translation only. ∎ The twin is a true twin (not S itself) unless S also has the mirror M as a symmetry. This is the spinel law of cubic crystals [9], where the twin is a half turn about ⟨111⟩; for a solid with the 120° turn the half turn and the 60° turn give the same twin.

The rule is checked exactly for the OG, the DICTO OTT, the DBT and the golden envelope (Table 1), with the rational 60° turn that takes (1,0,0) to (2/3, 2/3, −1/3); for the OCD Stella it was checked exactly in [3].

3. The fit rule

Rule. If a solid lies in its own layer, and its copies in the octahedra of one layer do not overlap, then it fits every stacking.

Why. Layers are disjoint slabs, so solids in different layers cannot overlap. Every layer of every stacking is the reference layer moved by a translation or by a mirror; a translated layer carries translated solids, and a mirrored layer carries the mirror images, which by the twin rule are the solids turned 60°. Within a layer nothing depends on the word. ∎ A solid inside its octahedron lies in its layer, so it fits every stacking with the cell kit of [3]: the octahedron cut into the solid and 8 caps by the planes through its centre and edges, the tetrahedra whole; at an h plane two caps join into a twin cap and two tetrahedra into a bipyramid.

Solid	Centre-symmetric	120° turn about (1,1,1)	Mirror across the layer	Mirror image = 60° turn	In its octahedron	In its layer	Fits	Share of space
OG	yes	yes (all four axes)	no	yes	yes	yes	every stacking	$1025/114 - 145\sqrt{5}/38 \approx 45.89\%$
OCD Stella	yes (float)	yes, all four axes (float)	no (float)	yes [3]	yes [3]	yes	every stacking	≈ 51.63% (float)
DICTO OTT	yes	yes (all four axes)	no	yes	yes	yes	every stacking	$176/3 - 26\sqrt{5} \approx 52.89\%$
DBT	yes	yes (this axis only)	no	yes	**no**	yes	**every stacking**	$493/6 - 73\sqrt{5}/2 \approx 55.02\%$
golden envelope	yes	yes	no	yes	no	**no**	the octet only	$90 - 40\sqrt{5} = 10/\varphi^6 \approx 55.73\%$ [2]

Table 1. Exact tests (polytype_exact.py) for one octahedron at the origin, except where marked float (the OCD Stella's symmetries and share, from the float copy of its exact corners; its twin and its place in the octahedron are exact in [3]). Shares are of the whole space and are the same in every stacking the solid fits.

The DBT. The DBT pokes out of its octahedron through its six side faces, into the tetrahedra of its own layer, but its two flat poles keep it inside the layer. In [3] it was shown to fill the gyrated octet's octahedra pole to pole without overlap; by the fit rule it fits every stacking. The exact overlap test agrees: with the reference DBT in the middle layer of three and every neighbour within reach, there is no overlap for any of the four pairs of boundaries cc, ch, hc and hh.

The golden envelope. Without its two pole cuts the DBT is the golden envelope of [2], a rhombic triacontahedron. Its two three-fold tips leave the layer. Across a c plane they meet tetrahedra and the envelope fits, as in the octet [2]; across an h plane the tips of two envelopes meet and the envelopes overlap, by exactly $(-88 + 40\sqrt{5})/81 \approx 0.017811$, which is the volume of the two tips the DBT cuts off. So the cut that makes the DBT is exactly what every stacking needs.

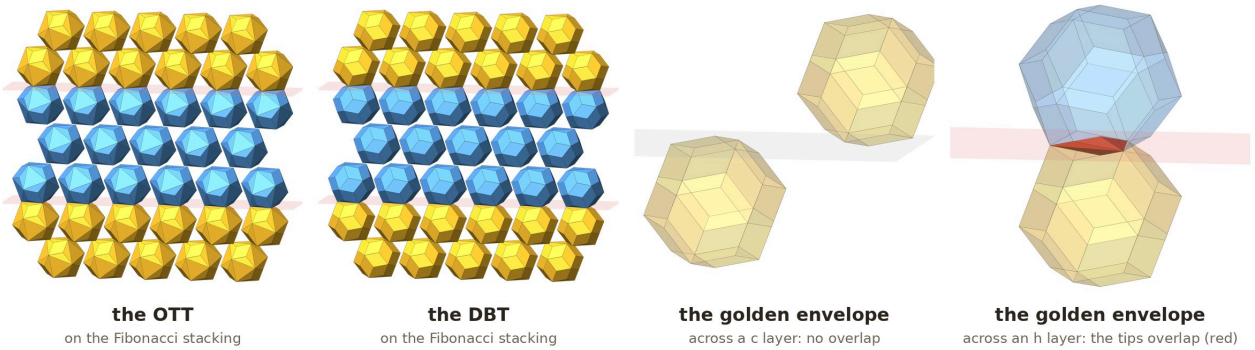


Figure 1. Left: the DICTO OTT and the DBT on the Fibonacci stacking (first orientation gold, mirror twins blue, h planes faint red). Right: two golden envelopes across a c plane (no overlap) and across an h plane, where their tips overlap (red).

4. Polytypes

For the OG, the pieces per cell follow from the share f of h planes. Each cell has one plane between its layer and the next, so per cell there are f twin caps and f bipyramids, and the caps and tetrahedra that did not join: $8 - 2f$ caps and $2 - 2f$ tetrahedra, with the OG (or its twin) once. The volumes are those of [3]: cap $(277 - 79\sqrt{5})/1539$, twin cap twice that, tetrahedron $(16 + 8\sqrt{5})/81$, bipyramid twice that. The twin-cap counts were also counted from the geometry, layer by layer (polytype_exact.py), for every word in Table 2.

Polytype	Word (Jagodzinski)	Packing	Share of h planes f	Caps	Twin caps	Tetrahedra	Bipyramids
3C (octet)	c	ABC	0	8	0	2	0
2H (gyrated octet)	h	AB	1	6	1	0	1
4H	hc	ABCB	1/2	7	1/2	1	1/2
6H	hcc	ABCACB	1/3	22/3	1/3	4/3	1/3
9R	hhc	ABACACBCB	2/3	20/3	2/3	2/3	2/3
15R	hcchc	ABCACBCABACABCB	2/5	36/5	2/5	6/5	2/5
Fibonacci	c → ch, h → c	never repeats	$(3 - \sqrt{5})/2$	$5 + \sqrt{5}$	$(3 - \sqrt{5})/2$	$\sqrt{5} - 1$	$(3 - \sqrt{5})/2$

Table 2. Kit pieces per cell for the OG, one OG (or twin) per cell in every case. Ramsdell symbols and packings computed from the words; 4H, 6H, 9R and 15R are among the best-known polytypes of silicon carbide and of close-packed metals [7, 8].

5. A Fibonacci crystal of OGs

Take the word grown by c → ch, h → c from c: c h c c h c h c c h c c h …. Its share of h tends to $1/\varphi^2 = (3 - \sqrt{5})/2$ (exactly 2584 of the first 6765 letters, both Fibonacci numbers), and since that number is irrational the word never repeats. No two h planes are adjacent and no three c planes follow each other, so the twin planes come 2 or 3 layers apart, in the Fibonacci order. By the fit rule the OG, the OTT, the OCD Stella and the DBT fill their places on this stacking exactly, and by Table 2 the OG crystal needs per cell $5 + \sqrt{5}$ caps, $(3 - \sqrt{5})/2$ twin caps, $\sqrt{5} - 1$ tetrahedra and $(3 - \sqrt{5})/2$ bipyramids: golden numbers, from the stacking alone. Each layer is periodic and the order of the layers is not, so the crystal is periodic in two directions and aperiodic in the third, with the same pieces as the octet and the gyrated octet. Fibonacci

words are the standard example of aperiodic order in one dimension [10]; Fibonacci chains have also been used to describe heavily faulted stackings of ZnS (web search, Farkas-Jahnke; IUCr congress abstract, not read in full).

6. Prior art and what is claimed

- **Stackings.** The two ways to lay one layer of the honeycomb on the next, and the infinitely many stackings this allows, are known [4, 5, 11]; the h/c description and polytype names are classical crystallography [6, 7, 8]; the spinel law is classical [9]. We claim none of these.
- **The twin rule and the fit rule** are short consequences of symmetry, and we expect that crystallographers would recognise them; we state them for solids placed in the honeycomb and did not find them stated in this form.
- **Claimed:** that the OG, the OCD Stella, the DICTO OTT and the DBT fit every stacking of the honeycomb, exactly, with the shares of space above (the DBT's fit beyond the gyrated octet is new here); that the golden envelope fails at h planes by exactly the volume the DBT removes; the exact OG kits for the polytypes; and the Fibonacci crystal of OGs with its golden piece counts.
- **A note on the DICTO Jewel.** [3] reports that the DICTO Jewel, framed by cubes, does not twin; that concerns its packing. As a solid the Jewel is pyritohedral, so the twin rule applies to it too; what fails (float, [3]) is the Jewel's packing with stella octangulas across a (111) plane, which is not a packing in the layers of this honeycomb.

In the sources checked (the full text of the 80 library sources listed in the appendix, and web searches on 20 October 2026) we found the stackings and the notation but not these solids on them.

7. Further work

- The space the DBT and the OTT leave on mixed stackings: the fill sets of [3] are for the octet and the gyrated octet.
- The OG family on other aperiodic stackings, and their diffraction.
- The polytypes in the Kaleidohedra app, with the word chosen by the user.

Data and reproducibility

Exact ($\mathbb{Q}(\sqrt{5})$): polytype_exact.py, with log polytype-exact.log (29 PASS) and data polytype-exact.json; it uses the exact corners of the OG ([2], studies/2026-10-18-ocd-octet/data/step1-ocd-exact.json), the DBT and the OTT ([3], dbt-exact.json, cut8-exact.json) and the exact tools of [2] (q5.py, exactpoly.py). Figures: polytype_figs.py (float copies of the exact solids), shots.mjs, polytype_compose.py.

References

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2. DICTO, The DICTO Ogdoecontahedron on the Octet Truss, DICTO DOI: 10.5281/zenodo.23457016.
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9. Crystal twinning (the spinel law), Wikipedia, https://en.wikipedia.org/wiki/Crystal_twinning (accessed 20 October 2026).
10. M. Baake, U. Grimm, Mathematical diffraction of aperiodic structures, Chem. Soc. Rev. 41, 6821–6843 (2012).
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Appendix. The reference library searched

The full text of the 80 sources below in DICTO's reference library was searched on 20 October 2026 for: polytype, stacking fault, stacking sequence, Jagodzinski, Ramsdell, hexagonal close packing, twin, spinel, Fibonacci and gyrated. None contains the results of this paper. The relevant hits are the Wikipedia pages on crystal twinning and on the honeycomb [4, 9], Cadotte et al. [5] and Gabbrielli et al. [11] on the honeycomb's stackings, Kléman (2013) on stacking faults in quasicrystals, and Baake and Grimm [10] and Kramer (2011) on Fibonacci order. Thirteen further items are page images without a text layer (the patents of Fuller, Snelson, Pearce, Baer, Hogan, Bergman and Lalvani, and Fedorov 1885).

Web searches (20 October 2026): polytypism and the Ramsdell and Jagodzinski notations (checked: [6, 7, 8]); Fibonacci and quasiperiodic stackings of close-packed layers; stackings of the tetrahedral–octahedral honeycomb.

Author(s)	Year	Title
Luca Pacioli, Leonardo da Vinci (ill.)	1509	Divina proportione
Albrecht Dürer	1525	Underweysung der Messung mit dem Zirckel und Richtscheyt
Wenzel Jamnitzer, Jost Amman (engr.)	1568	Perspectiva corporum regularium
Johannes Kepler	1596	Prodromus dissertationum cosmographicarum continens Mysterium cosmographicum
Johannes Kepler	1619	Harmonices mundi libri V
Euclid, T. L. Heath (tr.)	1908	The Thirteen Books of Euclid's Elements, Vol. III (Books X-XIII)
M. Goldberg	1937	A class of multi-symmetric polyhedra
Haresh Lalvani	1990	Transformational part-count in layered octahedral-tetrahedral truss configurations (NASA TM-102635)
P. W. Messer	1995	Stellations of the rhombic triacontahedron and beyond
M. Baake, F. Gähler	1998	Symmetry structure of the Elser-Sloane quasicrystal
T. C. Hales	1998	An overview of the Kepler conjecture
M. Baake	1999	A guide to mathematical quasicrystals
O. Ogievetsky, Z. Papadopolos	1999	On Quasiperiodic Space Tilings, Inflation and Dehn Invariants
U. Betke, M. Henk	1999	Densest lattice packings of 3-polytopes
Z. Papadopolos, O. Ogievetsky	1999	On Inflation Rules for Mosseri-Sadoc Tilings
Z. Papadopolos, O. Ogievetsky	1999	On quasiperiodic space tilings, inflation and Dehn invariants
R. V. Moody	2000	Model sets: a survey
M. Senechal	2004	Donald and the golden rhombohedra
S. Kabai	2007	54-Faced Space-Filling Polyhedron (Wolfram Demonstration; description and screenshot)
S. Kabai	2007	Golden Octet Truss (Wolfram Demonstration; description and screenshot)
Wikipedia contributors (2007 mirror)	2007	Kyushindo (2007 version)
S. Kabai	2008	A Rhombic Dodecahedron Made of Rhombic Triacontahedra (Wolfram Demonstration; description and screenshot)
S. Kabai	2008	Relating a Rhombic Triacontahedron and a Rhombic Dodecahedron (Wolfram Demonstration; description and screenshot)

M. Koca, M. Al-Ajmi, N. O. Koca	2009	Grand antiprism and quaternions
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Quaternionic representation of snub 24-cell and its dual polytope derived from E8 root system
S. Torquato, Y. Jiao	2009	Dense packings of polyhedra: Platonic and Archimedean solids
B. Grünbaum	2010	The Bilinski dodecahedron, and assorted parallelohedra, zonohedra, monohedra, isozonohedra and otherhedra
P. Kramer	2010	Gateways towards quasicrystals
A. M. Cutler, E. Schulte	2011	Regular polyhedra of index two, I
D. Frettlöh	2011	Icosahedral tilings in R3: the ABCK tilings (draft chapter)
J. Koskinen	2011	Unusual tilings and transformations
J. de Graaf, R. van Roij, M. Dijkstra	2011	Dense regular packings of irregular non-convex particles
M. Koca, N. O. Koca, M. Al-Ajmi	2011	Branching of the W(H4) polytopes and their dual polytopes under the Coxeter groups W(A4) and W(H3) represented by quaternions
M. Koca, N. O. Koca, M. Al-Barwani	2011	Snub 24-cell derived from the Coxeter-Weyl group W(D4)
T. Etzion	2011	Product constructions for perfect Lee codes
A. Garber, A. Gavrilyuk, A. Magazinov	2012	The Voronoi conjecture for parallelohedra with simply connected delta-surface
J. C. Lagarias, C. Zong	2012	Mysteries in packing regular tetrahedra
M. Baake, U. Grimm	2012	Mathematical diffraction of aperiodic structures
M. Koca, N. O. Koca, R. Koc	2012	Affine Coxeter group Wa(A4), quaternions, and decagonal quasicrystals
P. F. Damasceno, M. Engel, S. C. Glotzer	2012	Crystalline assemblies and densest packings of a family of truncated tetrahedra and the role of directional entropic forces
R. Gabbrielli, Y. Jiao, S. Torquato	2012	Families of tessellations of space by elementary polyhedra via retessellations of face-centered-cubic and related tilings
S. Torquato, Y. Jiao	2012	Organizing principles for dense packings of nonspherical hard particles: not all shapes are created equal
M. Kléman	2013	Defects in quasicrystals, revisited: I-flips, approximants, phason defects
N. Fujita, H. Takano, A. Yamamoto, A.-P. Tsai	2013	Cluster packing geometry for Al-based F-type icosahedral alloys
P.-P. Dechant, J. Wardman, T. Keef, R. Twarock	2014	Viruses and fullerenes: symmetry as a common thread?
A. Campello, G. C. Jorge, J. E. Strapasson, S. I. R. Costa	2015	Perfect codes in the lp metric
K. Deguchi, M. Nakayama, S. Matsukawa, K. Imura et al.	2015	Crystal structure of superconducting 1/1 cubic Au-Ge-Yb approximant with Tsai-type cluster
M. Baake, D. Damanik, U. Grimm	2015	What is aperiodic order?
M. Firsching	2015	Computing maximal copies of polytopes contained in a polytope
N. O. Koca, M. Koca, A. Al-Mukhaini, A. Al-Qanobi	2015	Quaternionic representations of the pyritohedral group, related polyhedra and lattices
R. Twarock, M. Valiunas, E. Zappa	2015	Orbits of crystallographic embedding of non-crystallographic groups and applications to virology
S. Kabai	2015	Cluster of 14 Rhombic Triacontahedra (Wolfram Demonstration; description and screenshot)
A. T. Cadotte, J. Dshemuchadse, P. F. Damasceno, R. S. Newman, S. C. Glotzer	2016	Self-assembly of a space-tessellating structure in the binary system of hard tetrahedra and octahedra
C. Hann, J. E. S. Socolar, P. J. Steinhardt	2016	Local growth of icosahedral quasicrystalline tilings
R. M. K. Dietl, J.-H. Eschenburg	2016	The icosahedral quasiperiodic tiling and its self-similarity
A. Stergiou	2018	Bootstrapping hypercubic and hypertetrahedral theories in three dimensions
A. Garber	2019	Voronoi conjecture for five-dimensional parallelohedra
D. Poland, S. Rychkov, A. Vichi	2019	The conformal bootstrap: theory, numerical techniques, and applications
R. Twarock, A. Luque	2019	Structural puzzles in virology solved with an overarching icosahedral design principle
T. Ishimasa	2019	Icosahedral quasicrystal, 1/1 and 2/1 approximants in Zn-based ternary alloys containing Au and Yb/Tb
A. Al-Siyabi, N. O. Koca, M. Koca	2020	Icosahedral polyhedra from D6 lattice and Danzer's ABCK tiling
M. Koca, R. Koc, N. O. Koca, A. Al-Siyabi	2020	Icosahedral tiling with dodecahedral structures
N. O. Koca, R. Koc, M. Koca, A. Al-Siyabi	2020	Dodecahedral Structures with Mosseri-Sadoc Tiles
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian et al.	2020	Bi-axial woven tiles: interlocking space-filling shapes based on symmetries of bi-axial weaving patterns
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian, M. Ebert, J. Cui, C. Fu, C. Starrett	2022	Geometrically interlocking space-filling tiling based on fabric weaves

B.-X. Lao, S. Rychkov	2023	3D Ising CFT and exact diagonalization on icosahedron: the power of conformal perturbation theory
D. Smith, J. S. Myers, C. S. Kaplan, C. Goodman-Strauss	2023	An aperiodic monotile
R. Akpanya et al.	2024	Topologically interlocking blocks inside the tetroctahedrille
I. J. Dejter, L. R. Fuentes, C. A. Araujo	2025	Perfect truncated-metric codes in lattice graphs and superlattice graphs
S. Rychkov	2025	Conformal bootstrap: from Polyakov to our times
M. Kovačević	2026	An isodiametric theorem and lattice diameter-perfect codes in A3
R. Al Raisi, N. O. Koca, M. Koca, R. Koc	2026	Modified Mosseri-Sadoc tiles from D6
Wikipedia contributors	2026	Crystal twinning (incl. the spinel law)
Wikipedia contributors	2026	Fullerene
Wikipedia contributors	2026	Goldberg polyhedron
Wikipedia contributors	2026	Kenshiro Abbe
Wikipedia contributors	2026	Kyūshindō
Wikipedia contributors	2026	Mitsusuke Harada
Wikipedia contributors	2026	n-flake (Sierpinski icosahedron)
Wikipedia contributors	2026	Tetrahedral-octahedral honeycomb (incl. the gyrated form)